Lab 5: Splines and 3D Parametric Curves

Computer Graphics Lab, Wolfram Mathematica

In[1]:=
```
SetOptions[EvaluationNotebook[], ShowCellLabel → True];
```

In[2]:=
```
SetOptions[{Plot, Plot3D, ContourPlot, DensityPlot, ParametricPlot,
    ParametricPlot3D, ListPlot, Graphics3D, ListPointPlot3D, RevolutionPlot3D},
  ImageSize → Small, BaseStyle → {16, FontFamily → "Times"}];
```

In[3]:=
```
(* PROBLEM 1 *)
```

In[4]:=
```
a = 4 / 25
```

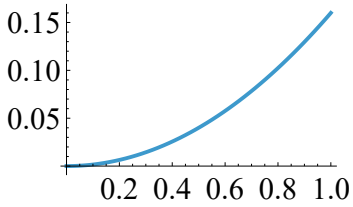
Out[4]=
$$\frac{4}{25}$$

In[5]:=
```
f = a #^2 &
```

Out[5]=
$$a\,\#1^2\,\&$$

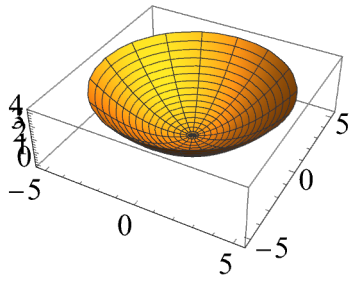
In[6]:=
```
f11 = Plot[f[x], {x, 0, 1}]
```

Out[6]=



In[7]:=
```
f14 = RevolutionPlot3D[f[x], {x, 0, 5}]
```

Out[7]=

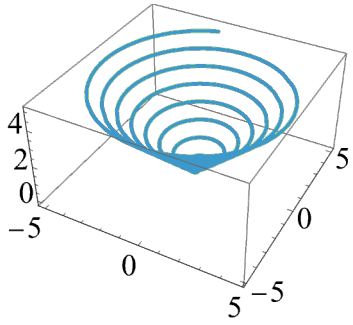


In[8]:=
```
R = 10
```

Out[8]=
```
10
```

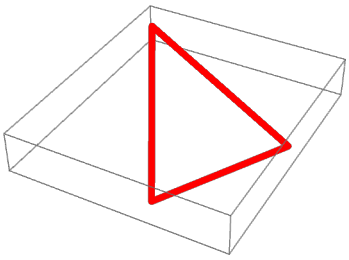
In[9]:=
```
f12 = ParametricPlot3D[{t * Sin[R * t], t * Cos[R * t], t}, {t, 0, 5}]
```

Out[9]=



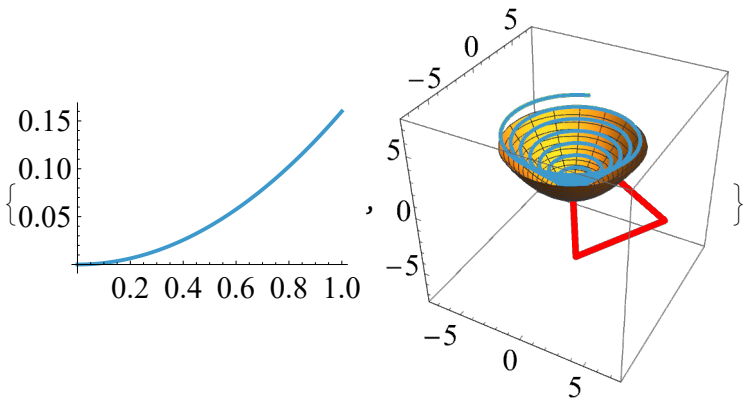
In[10]:=
```
f13 = Graphics3D[{Red, Thickness[0.02],
    Line[{{7, 0, 0}, {-7 / 2, 7 Sqrt[3] / 2, 0}, {7 / 2, -7 Sqrt[3] / 2, 0}, {7, 0, 0}}]}]
```

Out[10]=



In[11]:=
```
{f11, Show[{f12, f13, f14}, PlotRange → 7]}
```

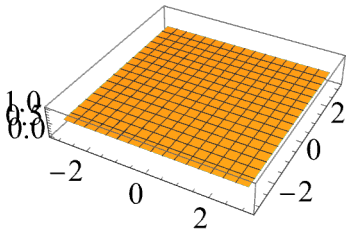
Out[11]=



In[12]:=
```
(* PROBLEM 2 *)
```

In[13]:=
```
G1 = ParametricPlot3D[{u, v, 0.5}, {u, -3, 3}, {v, -3, 3}]
```

Out[13]=

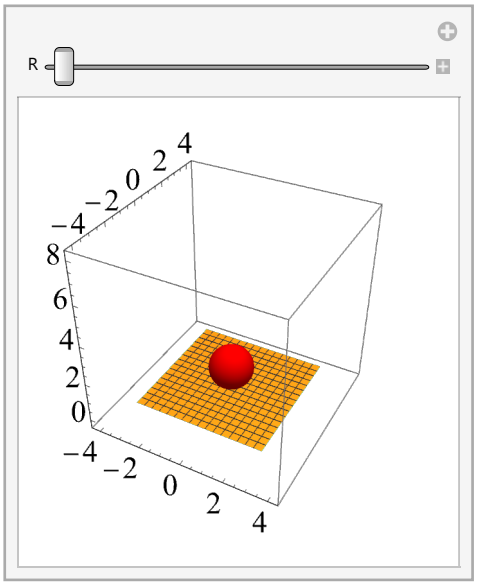


In[14]:=
```
cir[R_] := Graphics3D[{Red, Sphere[{0, 0, 0.5 + R}, R]}]
```

In[15]:=
```
All1[R_] := Show[{G1, cir[R]}, PlotRange → {{-4, 4}, {-4, 4}, {0, 8}}]
```

In[16]:=
```
Manipulate[All1[R], {R, 1, 3, 0.25}]
```

Out[16]=



In[17]:=
```
(* Problem 3 *)
```

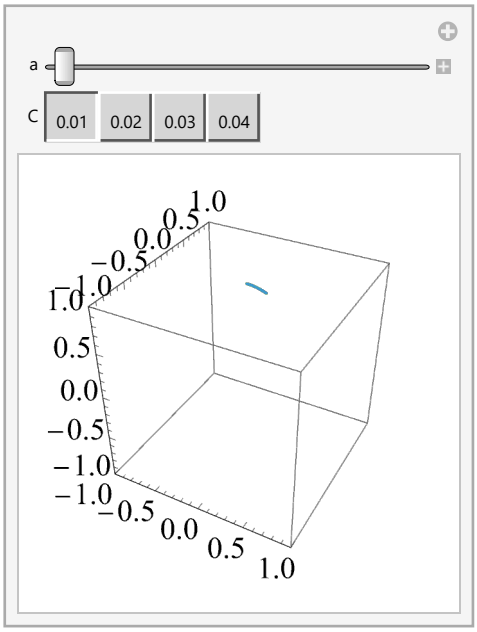
In[18]:=
```
p33[a_, C_] :=
 ParametricPlot3D[{Sin[2 t], 0, Cos[3 t]}, {t, 0, a}, PlotStyle → Thickness[C], PlotRange → 1]
```

In[19]:=
```
TList = {0.01, 0.02, 0.03, 0.04}
```

Out[19]=
```
{0.01, 0.02, 0.03, 0.04}
```

In[20]:=
```
Manipulate[p33[a, C], {a, 0.1, 2 Pi, 0.1}, {C, TList}]
```

Out[20]=



In[21]:=
```
(* Problem 1 *)
```

In[22]:=
```
data11 = {{1, 1}, {-2, 3}, {-1, -3}, {7, 0}, {3, 1}, {2, 2}}
```

Out[22]=
```
{{1, 1}, {-2, 3}, {-1, -3}, {7, 0}, {3, 1}, {2, 2}}
```

In[23]:=
```
f11 := Interpolation[data11, Method → "Spline"]
```

In[24]:=
```
f111 = Interpolation[data11, Method → "Spline"]
```

Out[24]=
```
InterpolatingFunction[ Domain: {{-2, 7}}
                       Output: scalar ]
```

In[25]:=
```
minx = Min[Transpose[data11][[1]]]
```

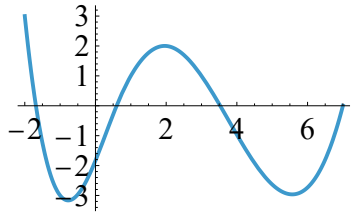
Out[25]=
```
-2
```

In[26]:=
```
maxx = Max[Transpose[data11][[1]]]
```

Out[26]=
```
7
```

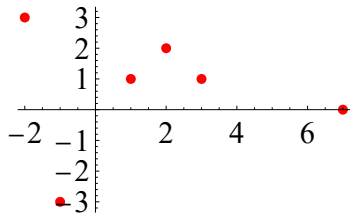
In[27]:=
```
plot11G = Plot[f11[x], {x, minx, maxx}]
```

Out[27]=



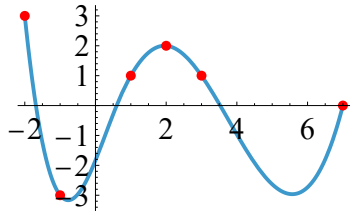
In[28]:=
```
plot11P = ListPlot[data11, PlotStyle → {PointSize[0.03], Red}]
```

Out[28]=



In[29]:=
```
Show[plot11G, plot11P]
```

Out[29]=



In[30]:=
```
(* Problem 2 *)
```

In[31]:=
```
x12 = Transpose[data11][[1]]
```

Out[31]=
```
{1, -2, -1, 7, 3, 2}
```

In[32]:=
```
y12 = Transpose[data11][[2]]
```

Out[32]=
```
{1, 3, -3, 0, 1, 2}
```

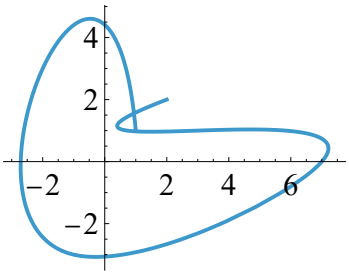
In[33]:=
```
sx12 := Interpolation[x12, Method → "Spline"]
```

In[34]:=
```
sy12 := Interpolation[y12, Method → "Spline"]
```

In[35]:=
```
spline12[t_] := {sx12[t], sy12[t]}
```

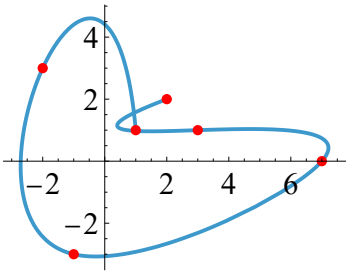
In[36]:=
```
plot12 = ParametricPlot[spline12[t], {t, 1, Length[x12]}]
```

Out[36]=



In[37]:=
```
Show[plot12, plot11P]
```

Out[37]=



In[38]:=
```
(* Problem 3 *)
```

In[39]:=
```
data3 = {{-3, -1}, {-1, -2}, {1, -2}, {2, -3}, {1, -4}, {-1, -4}, {-2, -5},
    {-2, -6}, {-1, -6}, {-1, -5}, {2, -5}, {3, -4}, {6, -5}, {7, -7}, {6, -8},
    {6, -9}, {7, -9}, {7, -8}, {8, -7}, {7, -4}, {6, -4}, {6, -2}, {5, 0}, {10, -1},
    {7, 0}, {10, 0}, {7, 1}, {10, 2}, {8, 2}, {10, 3}, {6, 2}, {5, 0}, {3, 0}, {1, 2},
    {1, 3}, {-2, 6}, {-3, 6}, {-4, 7}, {-4, 6}, {-6, 4}, {-6, 3}, {-5, 3}, {-3, 4},
    {-2, 2}, {-3, 1}, {-5, 2}, {-7, 0}, {-7, -1}, {-6, -1}, {-6, 0}, {-5, 1},
    {-4, 0}, {-5, -2}, {-3, -4}, {-2, -4}, {-2, -3}, {-3, -3}, {-4, -2}, {-3, -1}};
```

In[40]:=
```
x31 = Transpose[data3][[1]]
```

Out[40]=
```
{-3, -1, 1, 2, 1, -1, -2, -2, -1, -1, 2, 3, 6, 7, 6, 6, 7, 7, 8,
 7, 6, 6, 5, 10, 7, 10, 7, 10, 8, 10, 6, 5, 3, 1, 1, -2, -3, -4, -4, -6, -6,
 -5, -3, -2, -3, -5, -7, -7, -6, -6, -5, -4, -5, -3, -2, -2, -3, -4, -3}
```

In[41]:=
```
y31 = Transpose[data3][[2]]
```

Out[41]=
```
{-1, -2, -2, -3, -4, -4, -5, -6, -6, -5, -5, -4, -5, -7, -8,
 -9, -9, -8, -7, -4, -4, -2, 0, -1, 0, 0, 1, 2, 2, 3, 2, 0, 0, 2, 3, 6, 6,
 7, 6, 4, 3, 3, 4, 2, 1, 2, 0, -1, -1, 0, 1, 0, -2, -4, -4, -3, -3, -2, -1}
```

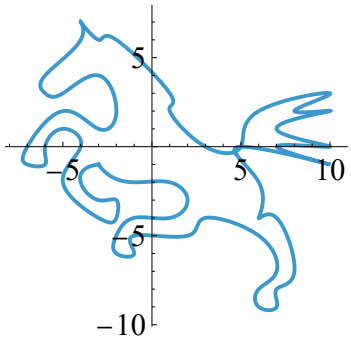
In[42]:=
```
sx31 := Interpolation[x31, Method → "Spline"]
```

In[43]:=
```
sy31 := Interpolation[y31, Method → "Spline"]
```

In[44]:=
```
spline31[t_] := {sx31[t], sy31[t]}
```

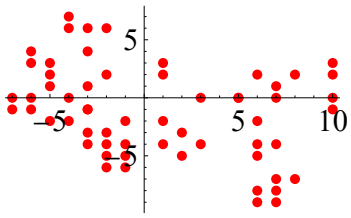
In[45]:= `plot31 = ParametricPlot[spline31[t], {t, 1, Length[x31]}]`

Out[45]=



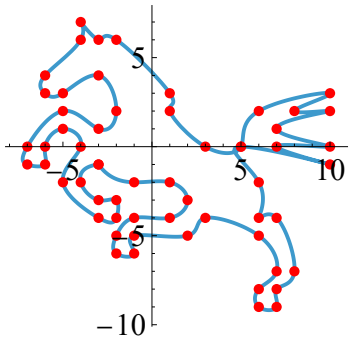
In[46]:= `plot32 = ListPlot[data3, PlotStyle → {PointSize[0.03], Red}]`

Out[46]=



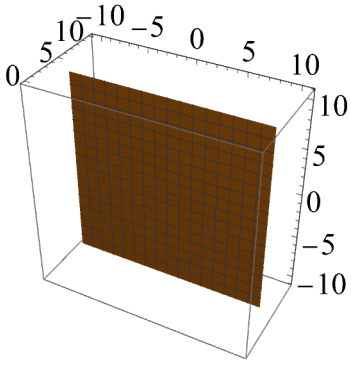
In[47]:= `Show[{plot31, plot32}]`

Out[47]=



In[48]:= `Plane3 = ParametricPlot3D[{x, 5, z}, {x, -10, 10}, {z, -10, 10}]`

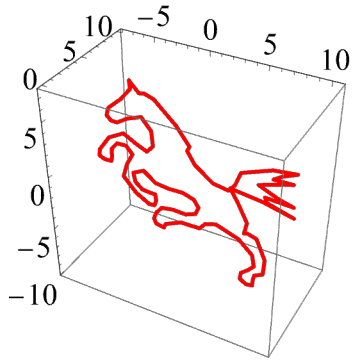
Out[48]=



In[49]:= `s3D[t_] := {sx31[t], 5, sy31[t]}`

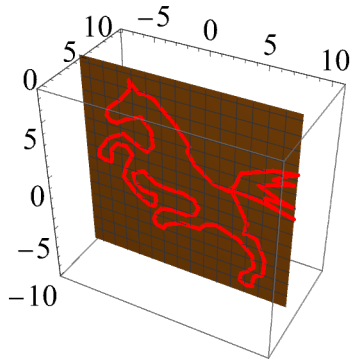
In[50]:=
```
Horse3 = ParametricPlot3D[s3D[t], {t, 1, Length[x31]}, PlotStyle → Red]
```

Out[50]=



In[51]:=
```
VerticalHorse = Show[Horse3, Plane3]
```

Out[51]=



In[52]:=
```
(* Problem 4 *)
```

In[53]:=
```
data14 = {{1, 1, 1}, {2, 3, 1}, {1, 3, 2}, {4, 3, 3}, {4, 4, 4}, {5, 3, 2}, {7, 5, 6}, {5, 5, 5}}
```

Out[53]=
```
{{1, 1, 1}, {2, 3, 1}, {1, 3, 2}, {4, 3, 3}, {4, 4, 4}, {5, 3, 2}, {7, 5, 6}, {5, 5, 5}}
```

In[54]:=
```
x14 = Transpose[data14][[1]]
```

Out[54]=
```
{1, 2, 1, 4, 4, 5, 7, 5}
```

In[55]:=
```
y14 = Transpose[data14][[2]]
```

Out[55]=
```
{1, 3, 3, 3, 4, 3, 5, 5}
```

In[56]:=
```
z14 = Transpose[data14][[3]]
```

Out[56]=
```
{1, 1, 2, 3, 4, 2, 6, 5}
```

In[57]:=
```
sx14 = Interpolation[x14]
```

Out[57]=

InterpolatingFunction [ Domain: {{1, 8}} Output: scalar]

In[58]:= **sy14 = Interpolation[y14]**

Out[58]=

InterpolatingFunction [ Domain: {{1, 8}} Output: scalar]

In[59]:= **sz14 = Interpolation[z14]**

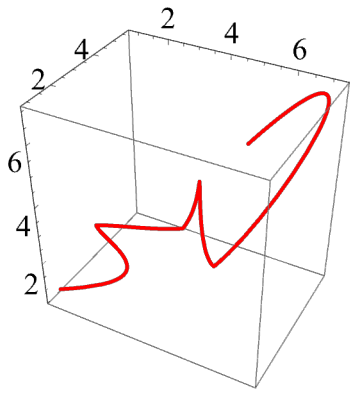
Out[59]=

InterpolatingFunction [ Domain: {{1, 8}} Output: scalar]

In[60]:= **s14[t_] := {sx14[t], sy14[t], sz14[t]}**

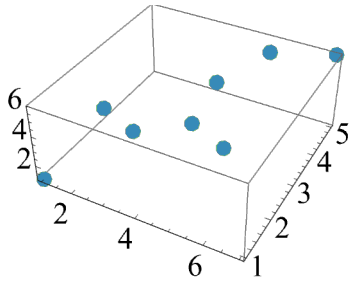
In[61]:= **k14 = ParametricPlot3D[s14[t], {t, 1, Length[x14]}, PlotStyle → Red]**

Out[61]=



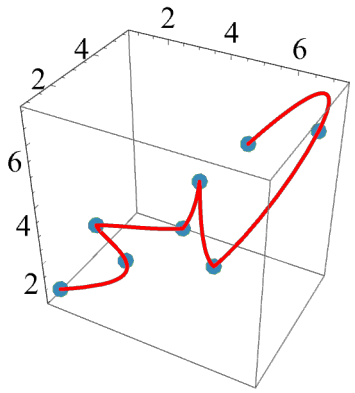
In[62]:= **plot14P = ListPointPlot3D[data14, PlotStyle → PointSize[0.05]]**

Out[62]=



In[63]:=
```
Show[k14, plot14P]
```

Out[63]=

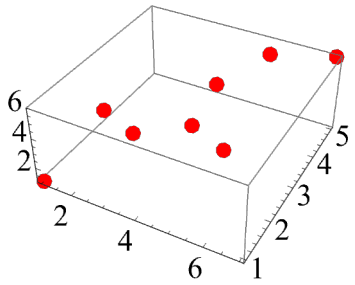


In[64]:=
```
(* Problem 5 *)
```

In[65]:=
```
plot5 := ParametricPlot3D[s14[t], {t, 1, Length[x14]}, PlotStyle → {clr5, Thickness[th5]}]
```

In[66]:=
```
plot5P = ListPointPlot3D[data14, PlotStyle → {PointSize[0.05], Red}]
```

Out[66]=



In[67]:=
```
plot5All := Show[{plot5, plot5P}, ImageSize → Small]
```

In[68]:=
```
MyMenu5 =
 PopupMenu[Dynamic[clr5], {Red → "Red", Green → "Green", Blue → "Blue", Purple → "Purple"}]
```

Out[68]=



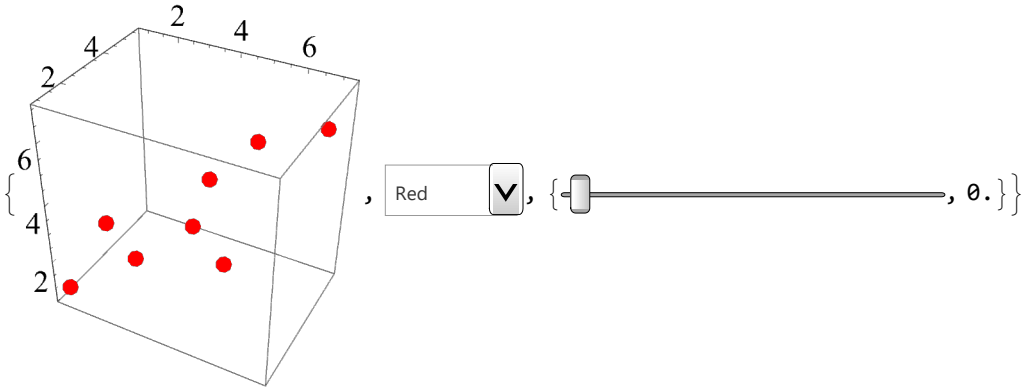
In[69]:=
```
MySlider5 = {Slider[Dynamic[th5], {0, 0.025}], Dynamic[th5]}
```

Out[69]=



In[70]:=
```
{Dynamic[plot5All], MyMenu5, MySlider5}
```

Out[70]=



In[71]:=
```
(* Bonus *)
```

In[72]:=
```
R = 2.5
```

Out[72]=
```
2.5
```

In[73]:=
```
xx[t_] := R * Cos[t]
```

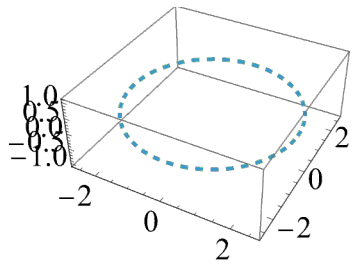
In[74]:=
```
yy[t_] := R * Sin[t]
```

In[75]:=
```
GLine1[t_] := Graphics3D[{Red, {Arrowheads[0.1], Arrow[{{0, 0, 0}, {xx[t], yy[t], 0}}]}}]
```

In[76]:=
```
GLine2[t_] :=
 Graphics3D[{Blue, {Arrowheads[0.1], Arrow[{{0, 0, 0}, {xx[t + Pi], yy[t + Pi], 0}}]}}]
```

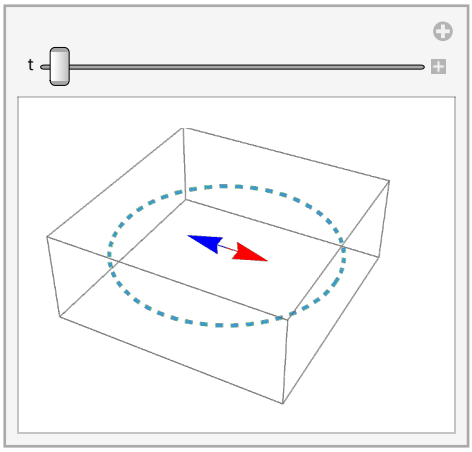
In[77]:=
```
Cirt = ParametricPlot3D[{xx[t], yy[t], 0}, {t, 0, 2 Pi}, PlotStyle → Dashed]
```

Out[77]=



In[78]:=
```
Manipulate[Show[{GLine1[t], GLine2[t], Cirt}], {t, 0, 2 Pi, Pi / 24}]
```

Out[78]=



In[79]:=
```
Graphics3D[{Arrowheads[0.1], Arrow[{{0, 0, 0}, {1, 1, 0}}]}];
```

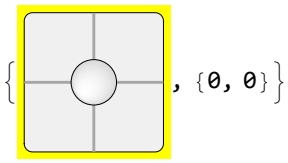
In[80]:=
```
(* Bonus 1 *)
```

In[81]:=
```
x1[t_, x_, R_] := t * Sin[R * t] + x
y1[t_, y_, R_] := t * Cos[R * t] + y
z1[t_] := t^2 / 10
```

In[84]:=
```
F1[x_, y_, t_, R_] := {x1[t, x, R], y1[t, y, R], z1[t]}
```

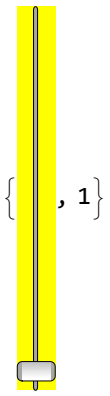
In[85]:=
```
S2 = {Slider2D[Dynamic[z], {{-15, -15}, {15, 15}},
    Background → Yellow, ImageSize → Large], Dynamic[z]}
```

Out[85]=



In[86]:=
```
SR = {VerticalSlider[Dynamic[R], {1, 5}, Background → Yellow], Dynamic[R]}
```

Out[86]=



In[87]:=
```
z = {0, 0}; R = 1;
```

In[88]:=
```
Clear[CenterS]
```

In[89]:=
```
CenterS[z_] := Graphics3D[{Red, PointSize[0.025], Point[{z[[1]], z[[2]], 0}]}]
```

In[90]:=
```
Clear[fb]
```

In[91]:=
```
fb[z_, R_] := ParametricPlot3D[F1[z[[1]], z[[2]], t, R],
  {t, 1, 15}, PlotRange → {{-20, 20}, {-20, 20}, {0, 30}}, PlotStyle → Black]
```

In[92]:=

In[93]:=
```
{Dynamic[Show[fb[z, R], CenterS[z]]], S2, SR}
```

Out[93]=

